

SatQuery AI - System Report

An agentic vision-language assistant for remote-sensing imagery. A controller reads the query and the input configuration, selects a specialist model, and returns an evidence-grounded answer with a visual overlay and an auditable execution trace.

1. System architecture

User query + image(s) > Controller (task + input routing) > Specialist model > Evidence + trace > Web UI

LAYER	TECHNOLOGY	STATUS
Frontend	React 19, Vite 8, Tailwind 4 (4 pages)	Built
Backend API	FastAPI, single-image endpoint	Built
Branch 1: single image	Qwen2-VL-7B-Instruct + LoRA	Not adapted
Branch 2: bi-temporal	ResNet-18 U-Net + rule engine	Measured
Branch 3: DeltaVLM	Adapter interface only (stubs)	Not built
Optical-SAR fusion	Not started	Not built

Branches share one contract (*ChangeEvidence*) so the controller consumes a single object shape regardless of which specialist answered.

2. Branch 1 - single-image VQA and captioning

ITEM	SPECIFICATION
Base model	Qwen2-VL-7B-Instruct (vision-language)
Adaptation	LoRA / PEFT, rank 16, alpha 32, q/k/v/o projections
Quantisation	4-bit (8 GB VRAM) or fp16 (16 GB)
Input	GeoTIFF, TIFF, PNG, JPEG via rasterio / PIL
Band handling	RGB by colour interpretation; Sentinel B04/B03/B02 by name; otherwise first three bands, recorded in the trace
Task routing	Keyword classifier -> captioning or VQA
Outputs	Answer or caption, model name, execution trace
Evaluation	RSVQA (exact match), VRSBench (BLEU, ROUGE-L, METEOR)
Measured accuracy	NOT YET MEASURED

The pipeline, LoRA configuration and evaluation scripts are complete, but no adapter has been trained and no benchmark has been run. The problem statement requires remote-sensing adaptation explicitly, so this is mandatory scope.

3. Branch 2 - bi-temporal change analysis

The published CDVQA baseline reaches about 58% because it never uses semantic labels. This branch predicts what each changed pixel was and became, then applies the benchmark's rules deterministically.

ITEM	SPECIFICATION
Architecture	U-Net, two decoders, ResNet-18 ImageNet encoder
Parameters	21.6 M
Input / output	6 channels (2 x RGB) at 512x512 -> two 7-class maps
Training	40 epochs, AMP fp16, RTX 4050 6 GB, best epoch 22
Answer engine	Deterministic rules -> one of CDVQA's 19 answer tokens
Inference	4 s model load, then about 200 ms per query
Interface	One call in, evidence object out; registry descriptor included
Tests	312 passing

Measured results

METRIC	VALUE	NOTE
CDVQA Test average accuracy	68.14%	968 scenes, 39,686 questions
CDVQA Test overall accuracy	74.89%	official held-out split
Published CDVQA baseline	~58%	we are about 10 points above
Majority-class floor	44.91%	what guessing alone scores
Rule ceiling (oracle)	99.90%	rules on ground-truth maps
Segmentation mIoU	42.08%	diagnostic, not the scored metric

The oracle figure proves the rules reproduce the benchmark's own answers almost exactly, so every remaining error is segmentation quality rather than reasoning. Near-duplicate screening bounds the true test figure at 67.7% or above.

4. Datasets

DATASET	ROLE	USED BY	SCALE
SECOND	Semantic change maps, training labels	Branch 2	4,662 pairs
CDVQA	Change VQA benchmark, 19 answers, 8 types	Branch 2	2,968 scenes
OSCD	Real Sentinel-2 13-band pairs, GeoTIFF path	Branch 2	24 pairs
BigEarthNet-S2	Domain adaptation corpus for LoRA	Branch 1	subset
RSVQA	Single-image VQA evaluation	Branch 1	not yet run
VRSBench	Captioning evaluation	Branch 1	not yet run

5. Problem statement coverage

MANDATORY REQUIREMENT	STATUS
Single-image VQA	Built, not measured
Second single-image task (captioning)	Built, not measured
Bi-temporal change analysis	Built and measured
Spatial change map	Built
Optical-SAR paired analysis	NOT BUILT
Remote-sensing adaptation (fine-tuning)	NOT DONE
Agentic tool orchestration	Partial
Confidence estimation	Built (branch 2, measured)
Visual evidence	Built (branch 2)
Execution trace	Built (both branches)
Downloadable report	Built (branch 2)
Interactive web application	Built

6. What distinguishes this build

- Confidence is measured, never invented. Every number is an accuracy actually observed on validation for that question type and land-cover class. Where no measurement applies it reports 0.0 and states why.
- Out-of-distribution input is detected. A physics-based producer cross-checks the trained model; when they disagree the system withdraws its confidence rather than answering confidently.
- Refusals over fabrication. SAR carries no land-cover mapping, so class questions are refused. Ungeoreferenced input reports pixel counts, never an invented area.
- Full execution trace: every stage records the tool, parameters, observation, duration and why the step was taken.
- Visual evidence is built from the same pixels the rule counted, so the picture cannot contradict the text.

7. Known gaps

- Remote-sensing fine-tuning on branch 1 has not been run. Mandatory.
- Optical-SAR paired analysis is not started. Mandatory.
- Branch 1 has no measured accuracy on any benchmark.
- Branch 2 under-predicts rare classes: water at 0.21x true frequency, playgrounds never predicted. Declared in the tool descriptor and reflected in its confidence.
- Branch 2 does not transfer across sensors yet: trained on sub-metre aerial imagery, it detects nothing at Sentinel-2's 10 m and correctly reports zero confidence.
- The controller registry spanning both branches is not yet wired.

All figures are measured on the stated splits. Nothing in this report is estimated or projected.